MILITARY SPECIFICATION

BLANKERS, INTERFERENCE, MX-8757/UPX AND MX-8758/UPX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the MX-8757/UPX and MX-8758/UPX interference blankers (defruiters) for the Identification Friend or Foe/Air Traffic Control Radar Beacon System (IFF/ATCRBS). It specifies the design and performance requirements for equipment which will eliminate (defruit) from received IFF video those pulses (fruit) which are not synchronous with the associated interrogation cycle, while passing to the decoding equipment only those replies which are determined to be synchronous.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment And Systems, Requirements For (Navy)
MIL-E-16400F	Electronic Equipment, Naval Ship And Shore: General Specification
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definition Of And Basic Requirements For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirementd For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-463	Definition And System Of Units, Electromagnetic Interference Technology
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification And Production Acceptance Test: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1130	Connection, Electrical, Solderless Wrapped

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DoD AIMS 65-1000	Technical Standard For The ATCRBS/IFF/MARK XII System (AIMS)

(Application for copies should be addressed to: Electronic Systems Division, Warner-Robins ALC (MMIME), Robins Air Force Base, GA 31098 (DoD AIMS Program Office).)

3. REQUIREMENTS

3.1 General. The requirements specified herein shall apply equally to the MX-8757/UPX and the MX-8758/UPX defruiting equipment as outlined herein, unless otherwise specified herein. The equipment shall be in accordance with MIL-E-16400F, except as hereinafter specified.

3.1.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.1.2 Definitions. Definitions of terms used in this specification are given in 6.4.

3.2 Equipment types and descriptions. The defruiting equipment shall be of two basic types as specified in 3.2.1 through 3.2.2.1.

3.2.1 MX-8757/UPX. The MX-8757/UPX defruiter shall be a pulse-to-pulse defruiter employing single defruiting only. Defruiting shall be on a single channel per mode basis.

3.2.1.1 Operational modularity. The design of the MX-8757/UPX defruiter shall be such that it shall be possible to add, delete, modify, extend, increase, or decrease the parameters of the defruiter detailed in 3.2.1.1.1 by the simple addition or deletion of printed circuit (PC) boards or modular plug-in subassemblies which may be accomplished at the field operating level.

3.2.1.1.1 Modular parameters. Modular parameters shall be as follows:

a. Modes — Either 1, 2, 3, or 4 mode configurations (any or all of modes 1, 2, 3/A, C, 4) with up to 2, 3, 4, or 5 modes (including mode 4) interlaced, respectively.

b. Range — Range shall be adjustable to 648 kilometers (km) (350 nautical miles (nm)), ±5 percent. The range requirement may be met by adding circuitry as needed to increase range in gross increments and use of internal switch or solderless link to adjust within each increment.

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3.2.2 MX-8758/UPX. The MX-8758/UPX defruiter design shall employ pulse-to-pulse single defruiting and shall perform up to four modes of defruiting (with up to five modes interlaced) using a single channel of memory. Bracket-defruiting shall be employed on the first interrogation cycle each time the mode of interrogation is changed.

3.2.2.1 Operational modularity. The design of the MX-8758/UPX defruiter shall be such that it shall be possible to increase or decrease the range of the defruiter in accordance with the requirements of 3.2.1.1.1.b by the simple addition or deletion of printed circuit boards or plug-in subassemblies which may be accomplished at the field operating level.

3.2.3 Reliability and maintainability. The reliability and maintainability requirements specified herein shall apply equally to the MX-8757/UPX and the MX-8758/UPX defruiters, unless otherwise specified herein. For the purposes of reliability and maintainability, both the MX-8757/UPX and the MX-8758/UPX equipments shall be units capable of four-mode, 648 km (350 nm) defruiting.

3.2.3.1 Reliability requirements.

3.2.3.1.1 Quantitative reliability. The defruiter shall have an upper test limit mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) of 2000 hours, and a lower test MTBF (θ_1 as defined in MIL-STD-781) of 1000 hours. Failure shall be the inability to conform to any of the performance, control, or monitoring functions specified herein.

3.2.3.1.2 Equipment conditioning. All equipments produced in accordance with this specification shall be capable of passing a temperature equipment conditioning test in accordance with 4.5.2.

3.2.3.2 Maintainability requirements.

3.2.3.2.1 Quantitative maintainability requirements. When repair is accomplished by replacement of subassemblies (modules or PC boards), the equipment shall have an equipment repair time (ERT) of no greater than 15 minutes. ERT includes localization, isolation, disassembly, interchange, reassembly, alignment, and check-out for all subassemblies and parts.

3.2.4 Painting. The paints and painting process shall be in accordance with the requirements of MIL-E-16400F.

3.2.5 Radio frequency interference. The components of the defruiter shall conform to the requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463, Class IC (where applicable) and the interference limits specified therein. In service, the defruiter will be operated near, or interconnected with, high power radar equipment from which high frequency energy will be radiated and conducted to the defruiter. Provisions shall be made by incorporation of shielding or filtering, or both, in the equipment for proper operation under such conditions as those specified in MIL-STD-461.

3.2.6 Environmental conditions. The equipment shall be capable of continuous operation under the conditions specified in MIL-E-16400F, modified as follows:

a. Temperature, Class 2 of MIL-E-16400F.
b. Ambient relative humidity, 0-95 percent.
c. Elevation, sea level to +3658 meters (m) (+12,000 feet (ft)).
d. Salt-laden air encountered in shipboard or coastal service.
e. Sand and dust particles 10 to 50 microns in size.

3.2.7 Storage, transportation and warm-up conditions. The equipment shall be capable of satisfactory operation after being stored for a period not exceeding one year at storage room ambient temperature ranges from -62° Celsius (C) to +75°C.

3.2.7.1 Warm-up time. The equipment shall be fully operable within 10 seconds from the non-operating condition without the use of heaters at ambient temperatures as specified in 3.2.6.

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3.2.7.2 Air transportation. The equipment shall be capable of air transportation at altitudes up to 15,240 m (50,000 ft) without damage, in non-pressurized, non-temperature controlled cargo aircraft for a flight time of 20 hours.

3.2.8 Ventilation. Ventilation may be provided by forced air. If required, blowers shall be an integral part of the defruiter. Blower motors shall not use brushes. Changes to blowers shall not be required as power input frequency is changed.

3.2.9 Composition. The MX-8757/UPX and MX-8758/UPX shall each consist of a single package.

3.2.10 Construction.

3.2.10.1 Modules. The requirements of MIL-E-16400F for modular construction are applicable. Materials and parts in accordance with MIL-E-16400F shall be used in the construction of this equipment; use of parts and materials not in full compliance is unacceptable, unless specifically approved by the procuring activity. Procedure I of MIL-STD-965 applies for non-standard parts. In keeping with design objectives of miniaturization and weight reduction, solid-state devices (of types capable of reliable performance at temperatures cited herein) shall be employed to the maximum extent practical.

3.2.10.1.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.2.10.2 Assemblies and subassemblies. As required by MIL-E-16400F, the equipment shall employ unitized construction and modular design to provide easily replaceable assemblies and plug-in subassemblies. Each assembly or subassembly shall provide a complete circuit function(s) with readily accessible test points at input and output areas. Plug-in subassemblies shall be of the printed circuitry type which can be removed without need of unsoldering and which can be easily repaired. Each PC board card or plug-in subassembly shall be individually keyed to prevent insertion into the wrong connector, or reversing.

3.2.10.3 Fabricated electronic modules (FEM) and standard electronic modules (SEM). The use of FEM or SEM is not required in the construction of this equipment.

3.2.10.4 Main chassis access. The main chassis shall be mounted to its external cabinet by channel slides. Suitable locking devices and cabling shall be provided so that the chassis may be fully extended from its cabinet for servicing. Complete removal of the chassis from its cabinet shall be possible without extensive disassembly.

3.2.11 Weights and dimensions. The maximum weight of the MX-8757/UPX and the MX-8758/UPX shall be 14.5 kilograms (kg) (32 pounds (lbs)) and 11.8 kg (26 lbs), respectively. The dimensions of the MX-8757/UPX shall be as follows:

a. Width (not including rack mounting brackets): 43.2 centimeters (cm) ±0.064 cm (17.00 inches (in) ±0.025 in)
b. Height: 12.7 cm ±0.064 cm (5.00 in ±0.025 in)
c. Depth (including front panel handles and rear panel connectors): 47.96 cm ±0.318 cm (18.88 in ±0.125 in).

The dimensions of the MX-8758/UPX shall be as follows:

d. Width: 43.2 cm ±0.064 cm (17.00 in ±0.025 in)
e. Height: 12.7 cm ±0.064 cm (5.00 in ±0.025 in)
f. Depth (including front panel handles and rear panel connectors): 42.88 cm ±0.318 cm (16.88 in ±0.125 in).

3.2.12 Electrical design. The electrical characteristics as required herein shall be achieved and maintained under any combination of service environmental and operating conditions. The equipment shall be designed and fabricated for a minimum loss of operational capability as functions are lost or become inoperative due to electrical, electronic, or physical failures.

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3.2.12.1 Primary power. The equipment shall be capable of operation from a supply voltage of 115 volts ±10 percent and supply frequencies of 50 hertz (Hz) ±10 percent, 60 Hz ±10 percent, and 400 Hz ±10 percent, single-phase. If a line voltage regulator is required, it shall be installed in the space allocated for the defruiter. The requirements of the Transient voltage, Transient frequency, and Pulsed loads paragraphs of MIL-E-16400F apply.

3.2.12.2 Overload protection. Overload protection shall be provided by fusing all primary lines of the alternating current (AC) power input as specified in MIL-E-16400F, and current limiting devices shall be provided on the power supply output. Blown fuse or limiter indicators are required and neon indicator lamps shall be used where practical. The type of fuseholder shall be the extractor post type in accordance with MIL-F-19207. Primary AC power fuses with indicators and associated spare fuses shall be mounted on the front panel of the equipment.

3.2.12.3 External connectors. The connectors shall be in accordance with MIL-E-16400F except mating connectors shall be furnished only for the primary power and function remoting connectors. All input and output connectors other than test connectors shall be located on the equipment's rear panel and (with the exception of primary power and function remoting connectors) shall be of the type C series on the MX-8758/UPX and type BNC on the MX-8757/UPX, or as specified in the contract (see 6.2.1). All connectors shall be provided with plastic protective caps. Identification of connector function shall be provided adjacent to each connector.

3.2.12.4 Connections for standard test equipment. Connectors shall be available for such external test equipment as may be required for operational check-out tests. Test connectors shall be the BNC type and shall be mounted on the front panel. Test connectors shall be provided with captive caps attached to the equipment for use when the connector is not in use.

3.2.12.5 Internal cabling/wiring. Internal cabling shall be in accordance with the requirements of MIL-E-16400F.

3.2.12.5.1 Automatic wire wrap. Wire wrapped connections, if used, shall be in accordance with MIL-STD-1130, Class A wraps. Conscious effort shall be made to use the minimum number of wires possible. Square wrap posts shall be used.

3.2.12.5.2 Point-to-point wiring. Point-to-point wiring shall not be utilized. Point-to-point wiring is going directly from one point to another by a straight or nearly straight line. Point-to-point wiring of adjacent terminals is allowed.

3.2.12.6 Fail-safe provisions. The design of the equipment shall be such as to be fail-safe; in the event of a failure or failures there shall not be additional or avalanche failures due to these failures. In addition, the design of the equipment shall be such as to preclude damage to the equipment due to incorrect or accidental turn-on or turn-off.

3.2.12.7 Test points. Both internal and external test points are required. Internal test points shall be probe type and shall be located at the top of each printed circuit card or subassembly and shall be easily identifiable and accessible. Signal inputs and outputs are an absolute minimum on all plug-in subassemblies. An external synchronizing trigger (BNC type connector) shall be provided on the front panel to allow monitoring of internal defruiting operation. Front panel test points should be probe type or BNC type. Probe type test points for all power supply voltages shall be provided both internally and externally.

3.2.13 Printed circuit boards. Where plug-in printed circuit boards are employed, they shall be in accordance with MIL-STD-275 and shall be readily removable from the chassis. Test points shall be easily accessible without withdrawing the board from its installed position. Board extenders allowing access to both sides of the board shall be provided to connect the printed circuit board into its normally connected circuitry for servicing purposes. Board extenders shall be stored within the equipment. Sufficient spacing shall be provided for adequate air circulation and to prevent accidental contact with adjacent cards during operation, testing and card extraction.

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3.2.13.1 Coating and treatments. Printed circuit boards shall be coated and treated in accordance with the requirements of MIL-E-16400F, except as otherwise specified herein. The minimum spacing between printed circuit conductors for uncoated boards shall be: 0.051 cm (0.020 in) for 0 to 13 volts direct current (DC) or AC peak, 0.064 cm (0.025 in) for 13 volts to 150 volts DC or AC peak, 0.127 cm (0.050 in) for 151 volts to 300 volts DC or AC peak, 0.254 cm (0.100 in) for 301 volts to 500 volts DC or AC peak, 0.0005 cm (0.0002 in) per volt for voltages greater than 500 volts DC or AC peak.

3.2.14 Tropicalization. Tropicalization shall not be achieved by use of varnish or other overcoating materials. The tropicalization requirements of MIL-E-16400F shall apply.

3.2.15 Interchangeability. The interchangeability requirement of MIL-E-16400F shall also apply to subassemblies and plug-in assemblies.

3.2.16 Test features. For test purposes, input and output BNC connectors for composite video and mode identification tags shall be located on the front panel. In STANDBY position, these connectors shall be electrically isolated from corresponding connectors on the rear panel and in this position, it shall be possible to both input these signals and to view their outputs on the front panel. In DEFRUIT position, it shall be possible by the front panel connectors to view rear panel inputs and outputs of the corresponding signals. Test signals fed into these front panel connectors during STANDBY or DEFRUIT positions shall not affect actual IFF signals being processed.

3.2.17 Elapsed time meter. An elapsed time meter shall be provided to indicate total operating time of the defruiter in hours up to 9999 hours. The type of elapsed time meter containing the element mercury is not allowed.

3.2.18 Terminal designation. Terminal identification shall be provided for all internal connectors (including terminal boards), and for all external connectors used to connect this equipment to other equipment. All external connectors connected to coaxial transmission lines shall be identified with the impedance of the transmission lines.

3.2.19 Shock, vibration, and inclination.

3.2.19.1 Shock. The equipment shall conform to the shock requirements of MIL-S-901, and shall be Grade A, Class 1, Type A on a lightweight machine.

3.2.19.2 Vibration. The equipment shall conform to the vibration requirements of MIL-E-16400F.

3.2.19.3 Inclination. The equipment shall conform to the inclination requirements of MIL-E-16400F.

3.2.20 Enclosure. The enclosure shall be drip-proof in accordance with MIL-STD-108.

3.2.21 Mounting. The MX-8757/UPX defruiter set shall be capable of being mounted in either a standard 48.26 cm (19 in) relay rack or on a platform. Carrying handles shall be provided. The MX-8758/UPX defruiter shall have a platform mount only.

3.2.22 Front panel controls. Tactile identification of front panel controls is not required.

3.2.23 Function identity plate. A plate shall be provided which shall identify the operational/functional capability of the defruiter. For instance, a typical marking may be (4 Mode) (350 nm) . This plate shall be removable or in some other manner alterable so as to allow change when the operational characteristics of the defruiter set are changed. Grease pencil marking or similar devices to accomplish this shall not be used.

3.2.24 Safety. The design of the equipment shall conform to Requirement 1 of MIL-STD-454.

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3.2.24.1 Leakage current. When practicable, the leakage current of the equipment shall not exceed 5 milliamperes (mA). On equipments which have an unavoidable leakage current in excess of 5 mA, a warning plate shall be attached to the front panel reading:

DANGER - DO NOT ENERGIZE OR CONNECT POWER
TO THIS EQUIPMENT UNLESS FRAME AND ALL
EXPOSED METAL PARTS ARE GROUNDED

3.3 Performance and detailed requirements.

3.3.1 Description. The equipment specified herein shall comprise a miniaturized video defruiter set, complete in accordance with all specification requirements, which shall operate in conjunction with the antenna, interrogator, processing, and display components of the shipboard or ground IFF systems.

3.3.1.1 Compatibility. The defruiting equipment shall be fully compatible when operated in connection with any IFF equipments conforming to the AIMS system requirements (AN/UPX-23 and AN/UPX-27). In addition, the defruiter equipment shall be capable of operation with the AN/UPX-1(), AN/UPX-6(), and AN/UPX-11() interrogator sets. It may be assumed by the contractor that the necessary mode, tags, reset signals, and so forth, will be available at the output of the interrogator sets.

3.3.2 Functional operation.

3.3.2.1 MX-8757/UPX. The total memory of the defruiter shall be functionally divided into subsections or channels; a separate memory channel for each different mode of interrogation is required. The size of each memory channel (or the number of bits) shall be directly proportional to the range of the defruiter and inversely proportional to the granularity (resolution) of each bit. When the defruiter set is initially turned on, the total memory shall be blank. The incoming video in response to the first interrogation is received from the interrogator set (or R/T unit) shall be processed (as required elsewhere in this specification), and shall be read into the appropriate mode memory channel. The mode of the incoming video is determined by the presence of the appropriate mode triggers or tags. The incoming video shall be clocked into memory until the internal defruiter clock has reached a count indicating that the selected range of the defruiter has been reached. When this occurs, clocking of incoming video into memory shall be terminated. When the video from the second interrogation on the same mode is presented to the defruiter (undelayed video), it shall be first processed and then read into the same memory channel in the same position and in the same manner as the previously stored reply video (delayed video). Approximately simultaneously with these operations, the delayed video shall be read out of this memory and shall be applied to the appropriate circuitry to develop the pulse acceptance gates. These pulse acceptance gates shall then be functionally ANDed with the incoming undelayed video to obtain the defruited video on a pulse-to-pulse single defruiting basis only. The defruited video (undelayed) shall then be processed (as required elsewhere in this specification) and provided to the output terminal(s) of the defruiter. The proper time relationship shall be maintained by circuitry internal to the defruiter. This process shall continue until the internal defruiter clock has reached a count indicating that the selected range of the defruiter has been reached. The process shall be repeated for the third, fourth, fifth, — — — nth reply on the same mode. Furthermore, the identical process shall be accomplished for each separate mode of interrogation. When a Mode 4 reply is presented to the defruiter it shall not be defruited but rather shall be recognized as a Mode 4 reply and automatically by-passed around the defruiter and presented to the output terminal(s) in the proper time relationship. The preceding narrative is the general functional operation of the MX-8757/UPX defruiter; however, no part of the preceding narrative shall be construed to take precedence over the requirements specified elsewhere in this specification.

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3.3.2.2 MX-8758/UPX. The MX-8758/UPX defruiter shall perform pulse-to-pulse single defruiting, employing only one memory channel. The same memory channel shall be used to accomplish single pulse-to-pulse defruiting for all modes of interogation. It is recognized that when operating in this manner every time the mode of interrogation is changed the defruiter will output erroneous information. Therefore, a means shall be provided to bracket-defruit on the first interrogation cycle each time the mode of interrogation is changed (with the exception of Mode 4 interrogations). Circuitry shall be incorporated in the mode recognition logic to recognize a change in interrogation mode. In all other respects, the operation of the MX-8758/UPX defruiter shall be identical to the operation of the MX-8757/UPX defruiter.

3.3.2.3 Performance function. The video defruiting equipment shall provide pulse-to-pulse single defruiting and bracket defruiting (as specified in 3.3.2.2) to perform the following functions:

a. Accept from an interrogation/receiver conforming to AIMS requirements (AN/UPX-23 and AN/UPX-27 as well as the AN/UPX-1( ), AN/UPX-6( ), and AN/UPX-11( ), mode video in the forms and formats outlined in 3.3 at any and all average duty factors (synchronous signals plus fruit and noise) up to 25 percent and pass to the decoder or processing equipment only mode reply pulses which are synchronous with the interrogation cycle.
b. Operate in the presence of continuous wave (CW) and modulated continuous wave (MCW) jamming. The use of circuits which impair equipment operation in the presence of railing and frequency modulation (FM) jamming shall be avoided whenever practicable. The anti-jamming circuits used shall not impair the normal operation of the equipment in the absence of jamming.
c. Provide an output signal which shall be full amplitude without countdown or deterioration of individual pulse characteristics for reply pulses whose leading edges do not change their positions relative to the last pulse (P_3) of the mode identification tags or relative to mode gates by more than 0.2 microseconds (μs) (including internal jitter of the defruiter) from one interrogation to the next on the same mode. The requirements of this paragraph shall apply in the presence or absence of fruit (non-synchronous replies) which precede or overlap the reply pulses.
d. Prohibit an output for a reply pulse whose leading edge changes its position relative to the last pulse (P_3) of the mode identification tags or leading edge of the mode gates by more than 0.80 μs $\pm$ 0.10 μs (including internal jitter of the defruiter) from one interrogation to the next on the same mode. For a reply pulse whose leading edge falls within the 0.80 μs $\pm$0.10 μs time position, only that portion of the pulse from the leading edge to the 0.80 μs $\pm$0.10 μs limit shall be passed. This requirement shall be checked for compliance with pulses of nominal pulse duration (0.45 μs).
e. Prohibit a pulse to the output of the defruiter whose pulsewidth is less than 0.3 μs
f. Automatically operate in synchronism with an interrogator with pulse repetition frequency (PRF) of up to 450 interrogations per second
g. Automatically operate in synchronism with an interrogator whose pulse repetition frequency constantly varies PRF limits 150 pulses per second (pps) to 450 pps
h. Pulse-to-pulse defruit to a range of up to 648 km (350 nm) up to four modes sequential (1, 2, 3/A, C) with up to five modes interlaced (1, 2, 3/A, C, 4)
i. Mode 4 replies shall not go through the normal defruiting process; instead a Mode 4 reply shall be detected at the input to the defruiter, automatically by-passed around the defruiter, and applied to the output terminal(s) in the proper time relationship. Pulsewidth discrimination shall not be applied to Mode 4 replies.

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3.3.2.4 Video input characteristics.

3.3.2.4.1 Video input formats. The video input to the defruiter may take anv one of the following forms:

a. Composite video with mode sensitive triggers for Modes 1, 2, 3/A, C, and 4 (comprising up to six input lines) for the MX-8757/UPX defruiter only
b. Separated video with mode identification tags and reset signals (comprising up to six input lines) for the MX-8757/UPX defruiters only
c. Composite video with mode identification tags and reset signals (comprising two input lines)
d. Tagged video (comprising one input line only). Tagged video shall use the composite video connector referenced in 3.3.2.4.1c.

3.3.2.4.2 Signal parameters. Input video to the defruiter shall consist of synchronous video signal combinations of any of the reply codes, non-synchronous signals (fruit) of any reply code combinations, random pulses, and noise. Synchronous video signals will normally have the following characteristics:

a. Positive polarity pulse trains
b. Peak voltage amplitude up to 6 volts limited at some level between 1.5 and 5 volts (limit normally adjusted to 2.0 volts)
c. Signal-to-noise ratio: Between 2.0 and 6.0. Minimum desirable signal (for) operation) 0.2V + 0.05V above noise
d. Amplitude variation. Less than 1 dB between any two pulses within a single pulse train
e. Spacing between first and last framing pulses of a train, 20.3 μs $\pm$ 0.1 μs
f. Pulse spacing within pulse train, in 1.45 μs $\pm$ 0.1μs increments from first pulse leading edge
g. SPI pulse: 4.35 μs from second framing pulse leading edge
h. Pulse duration, 0.45 μs $\pm$ 0.1 μs received at receiver input
i. Pulse rise time, 0.05 μs to 0.10 μs received at receiver input
j. Pulse decay time, 0.05 μs to 0.20 μs as received at receiver input
k. Pulse undershoot, less than five percent
l. Pulse position tolerances:
    1. Each pulse with respect to first bracket pulse $\pm$ 0.1 μs
    2. SPI pulse with respect to second bracket pulse $\pm$ 0.1 μs
    3. Any pulse with respect to another pulse of train $\pm$ 0.15 μs.

3.3.2.5 Trigger input. The defruiting equipment shall be capable of accepting tags, reset signals and mode gates in the form and formats specified in 3.3.2.4.1. Signal characteristics of the tags, reset signals and mode gates are as follows:

a. Polarity: Positive
b. Amplitude: 2 to 20 volts (at peak)
c. Duration: Tags and reset (see 6.4.7.1 and 6.4.8)
   Mode sensitive triggers (see 6.4.13)
d. Rise time: Less than 0.2 μs (10 percent to 90 percent point relative to peak amplitude)
e. Decay time: Less than 0.5 μs (90 percent to 10 percent point relative to peak amplitude)
f. Unintentional jitter between interrogator pulses of one interrogation cycle: Less than 0.1 μs
g. Pulse repetition frequency (PRF): From 0 to 450 interrogations per second
h. The mode trigger pulses may be in any or all of Modes 1, 2, 3/A, C, 4

3.3.2.5.1 Mode recognition circuitry. In order to minimize the possibility of false tags and resets, (fruit or jamming occuring on the video line which may be detected as false tags and resets) when operating in all input video formats except tagged video, the video shall not be routed through the mode recognition circuitry. When employing tagged video, the mode decoding circuitry shall have the capability to pulsewidth discriminate between video pulses which appear as false tags or resets, or both, and legitimate mode tags and resets.

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3.3.2.5.2 Trigger and gating circuitry. The circuitry in the defruiter shall be isolated from the input trigger terminals. The isolation shall be such that any signals generated within the defruiter which appear across the input terminals shall not exceed a level of 0.1 volt, peak-to-peak in the terminated or unterminated condition. The triggers shall be used as required to initiate sweep generation, select and process the mode of interrogation, initiate the gating for the read and write circuitry, and so forth. The delay of the start of the internal mode gate relative to the time of the receipt of P_3 at the defruiter mode trigger input connector shall not exceed 0.1 μs.

3.3.2.6 Internal video input impedance. All video inputs (mode video, mode identification tags, mode sensitive triggers, reset signals, and so forth) shall be terminated in 75 ohms $\pm$ 5 percent and shall be capacitively coupled with DC clamping to the internal circuit. By means of wire jumpers, the terminating resistors (located on an accessible terminal board) shall be wired in or out as an installation procedure. Unterminated video inputs shall be not less than 3000 ohms and not more than 50 micro-micro farads ($\mu\mu$F). When the defruiter is in the DEFRUIT position, the terminating resistors shall be connected across the video input terminals. When the defruiter is in STANDBY or OFF positions, the terminating resistors shall be disconnected from the video input and output terminals.

3.3.2.7 Signal-to-noise ratio. Types of circuits and video levels shall be chosen to minimize deterioration of signal-to-noise ratios. The defruiter shall have full output corresponding to synchronous signals for the defruiting operation when the input signal is 0.2 V $\pm$ 0.05 V above the noise.

3.3.2.8 Basic memory devices and resolution. The basic memory devices shall be either integrated circuit shift registers or magnetic ferrite core memory. The memory devices with associated logic and timing devices shall be such as to permit storing for periods up to 10 seconds all IFF interrogations (up to 450 interrogations per second) whose pulses are as small as 0.35 μs in width and whose pulse resolution is as specified in 3.3.2.3.

3.3.2.9 Range operation. The defruiter shall be designed so that when the desired range to which the defruiter is internally set is reached, the defruiter shall terminate defruiting operations for that particular interrogation and reset itself for the next cycle.

3.3.2.9.1 Adjustable range. The range shall be internally adjustable in increments of 40 km (25 miles) or less to 648 km (350 nm) $\pm$ 5 percent. Switches or solderless links shall be used for range selection. Range markings shall be provided to identify range increments.

3.3.2.10 Bandwidth. The defruiter circuitry shall have a bandwidth such as to minimize the deterioration of the signal-to-noise ratio of the overall defruiting equipment and to conform to the acceptance/rejection limits specified herein.

3.3.2.11 Threshold control. It is desirable for the defruiter to operate with a minimum discernible signal at all times. (Standard video peak can be expected to vary from 0.2 V to 5 V; normal video is 2 V peak). Under noise conditions, it is a requirement that the defruiter function as far down into the noise as practicable. Circuitry shall be incorporated in the defruiter to provide essentially optimum performance automatically under varying service conditions of signal and noise. This thresholding circuit shall be used to both amplitude and pulsewidth discriminate. Pulsewidth discriminator shall prevent narrow pulses (0.25 μs (- 0 percent, + 10 percent) at half voltage point) from entering memory resulting in acceptance gate generation.

3.3.2.11.1 Regenerative circuits. Regenerative type circuits shall not be used in the video chain prior to extraction of the signal from the noise.

3.3.2.12 Signal acceptance and rejection characteristics. The performance requirements specified in 3.3.2.12.1 through 3.3.2.12.2 shall be met for pulses of a nominal 0.45 μs width.

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3.3.2.12.1 Single source signals.

3.3.2.12.1.1 Synchronous signals. The output signal shall be full amplitude without countdown or deterioration of individual pulse characteristics for reply pulses whose leading edges do not change their positions relative to the last pulse (P_3) of the mode identification tags or relative to mode gates by more than 0.2 μs from one interrogation to the next on the same mode.

3.3.2.12.1.2 Non-synchronous signals. There shall be no output for a reply pulse whose leading edge changes its position relative to the last pulse (P_3) of the mode identification tags or leading edge of the mode gates by more than 0.80 μs $\pm$ 0.10 μs from one interrogation to the next on the same mode. For a reply pulse whose leading edge falls within the 0.80 μs $\pm$ 0.10 μs time position, only that portion of the pulse from the leading edge to the 0.80 μs $\pm$ 0.10 μs limit shall be passed.

3.3.2.12.2 Rejection characteristics for three non-synchronous input sources. When three non-synchronous prime number frequency signal sources are connected to the input circuit through a non-additive mixer, the maximum number N of output pulses shall be as calculated from the following formula:

For single defruiting operation:

$$N = 6m^2 (W_1 + W_2 - 2d)$$

Where
- W_1 = Width of pulses in the undelayed channel in seconds
- W_2 = Width of pulses in the delayed channel in seconds
- d = Overlap in seconds of delayed and undelayed pulses required to produce a full amplitude output
- m = A very close approximation of the mean pulse repetition frequency of the three sources in pulses per second

Based on this formula, with three completely non-synchronous sources connected to the input circuit, the number of output pulses shall not exceed the following table when W_1 = 0.5 μs, W_2 = 1.0 μs, and d = 0.2 μs.

Mean PRF of sources	Maximum number of output pulses, per second
200 PPS	1.0
2400 PPS	40.
5000 PPS	170.

3.3.2.13 Video output characteristics. The video output shall be provided in the following forms:

a. When the input video is as specified in 3.3.2.4.1a, the output video shall consist of composite video (comprising one output line) for MX-8757/UPX defruiters only.
b. When the input video is as specified in 3.3.2.4.1b, the output video shall consist of at least a separated video with composite tags and reset signals (comprising up to six output lines) for the MX-8757/UPX defruiter only.
c. When the input is as specified in 3.3.2.4.1c, the output video shall consist of at least composite video with composite mode tags (comprising two output lines). When operating in this manner, two electrically isolated video outputs shall be provided for the composite video output as well as the composite tags and reset signals.

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d. When the input video is as specified in 3.3.2.4.1d, the output video shall consist of at least tagged video output (comprising one output line only). When operating in this manner, two electrically isolated video outputs shall be provided for the tagged video output. These connectors shall be the composite video connectors referenced in 3.3.2.13c.
e. A separate isolated Mode 4 video output line shall be provided for the MX-8758/UPX defruiter in addition to its specified outputs.
f. The mode sensitive triggers and mode identification tags and reset signals shall be mixed as appropriate with the output defruited video and the undefruited mode 4 video and shall be provided to the output connectors in the proper format and time relationship so that the output video from the defruiter (with appropriate mode triggers, tags and reset signals) looks identical (within the specified tolerances) to the input video except that modes 1, 2, 3/A, and C are defruited.

3.3.2.13.1 Narrow pulse rejection. A pulsewidth discrimination circuit shall be employed in the final stages of the defruiter (when defruiting operation is selected) to prevent video pulses of less than 0.30 μs (- 0 percent, + 10 percent) duration (at half voltage points) from being passed to the video output circuitry.

3.3.2.13.2 Signal characteristics. The video outputs related to specific interrogation modes shall be regulated to provide an output only during the reply period related to that interrogation mode.

3.3.2.13.2.1 OFF and STANDBY conditions.

3.3.2.13.2.1.1 Video pulses. Video signal characteristics as specified in 3.3.2.4.2 shall apply.

3.3.2.13.2.1.2 Mode identification tags, reset signals and mode sensitive triggers. Signal characteristics shall be as specified in 3.3.2.5.

3.3.2.13.2.2 Defruit condition.

3.3.2.13.2.2.1 Video pulses.

a. Polarity: Positive
b. Amplitude: Adjustable by output amplifier gain control to between 1.0 and 5.0 volts $\pm$ 5 percent (normally adjusted to 2.0 volts)
c. Rise time: Input rise time $\pm$ 10 percent
d. Fall time: Input fall time $\pm$ 10 percent
e. Duration: Same as input video pulse (undelayed video pulse) $\pm$ 0.3 μs (with exception specified in 3.3.2.13.1)
f. Pulse spacing: Spacing of any pulse referenced to start pulse of the output train shall not differ from that of the input train by more than $\pm$ 0.02 μs.
g. Amplitude droop or overshoot: Less than 1 dB
h. Amplitude variation between any two pulses: Less than $\pm$ 5 percent
i. Humps, ringing or baseline modulation: Less than 0.05 V peak-to-peak

3.3.2.13.2.2.2 Mode identification, tags, reset signals, and mode sensitive triggers. Characteristics are as specified in 3.3.2.5.

3.3.2.13.3 Cross-coupling and other undesired outputs. Isolation between mode modules, filtering, and spurious outputs shall be such that all outputs, except those due to desired coincidence, will be at least 30 dB below the normal output under the desired coincidence conditons. Switching transients and similar effects shall be maintained at such levels that flashing, strobing, or baseline shift on an associated radar display unit does not occur as a result of the operation of any portion of the equipment provided in accordance with this specification.

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3.3.2.14 Internal video output impedance. All video outputs (mode video, mode identification tags, reset signal, and so forth) shall be DC coupled and shall match the characteristic impedance of a nominal terminated 75 ohm coaxial cable.

3.3.2.15 Overall video delay. The effective overall delay introduced into the IFF system by the video defruiting equipment shall be the minimum practicable value consistent with the required performance. Delay between either the OFF or STANDBY condition and the DEFRUIT condition shall not exceed 0.8 μs for both the MX-8757/UPX and the MX-8758/UPX. Any delay network required shall be of adequate quality to conform to the rise time and pulse stretching limitations specified herein and shall have an insertion loss of less than 0.5 dB.

3.3.2.16 Controls. In addition to the controls specified elsewhere in this specification, the defruiter shall have the controls and capabilities specified in 3.3.2.16.1 through 3.3.2.16.3.1.

3.3.2.16.1 General. Each control circuit of the defruiting equipment shall be designed to be operated by the application of a ground connection to a terminal on the defruiter equipment. All controls located on the front panel shall be identified. Those infrequently used within the chassis shall be screwdriver adjusted.

3.3.2.16.2 ON-OFF. To provide for control of the input power, an ON-OFF switch will be provided on the front of the equipment. When this switch is in the OFF position, no primary power shall be applied to any of the defruiting equipment circuits and the video input terminals shall be connected directly to the video output terminals through any passive delay network required to conform to the delay stability requirements. A light indicator shall be provided adjacent to this switch to indicate its state.

3.3.2.16.3 STANDBY-DEFRUIT-REMOTE. A three position switch shall be provided to select the STANDBY, DEFRUIT, or REMOTE conditions. When in the STANDBY condition the input video shall be connected to the defruiter output as outlined in 3.3.2.16.2. When in the DEFRUIT condition, the defruiter shall assume normal defruiting operation. A time delay relay or similiar device shall be interconnected with this switch (if required) to prevent damage to the equipment or false information passing out of the defruiter until all circuits have had sufficient time to warm-up and assume normal operation. When in the REMOTE position, remote control of the STANDBY-DEFRUIT condition shall be enabled to allow selection of these conditions from a remote position. Light indicators shall be provided adjacent to this switch to indicate whether the defruiter is in the STANDBY or DEFRUIT condition (selected either locally or remotely).

3.3.2.16.3.1 Remote control of defruiter state. A cable connector shall be provided on the rear of the defruiter cabinet and appropriately connected to allow control of the STANDBY and DEFRUIT functions from a remote position (as specified in 3.3.2.16.3). In addition, lead connections shall be made to this remoting connector to provide external, remote light indicators to indicate the following conditions: a) Grounding of one lead shall indicate the defruiter is in the STANDBY condition (when controlled either locally or remotely); b) Grounding of a second lead shall indicate the defruiter is in the DEFRUIT condition (when controlled either locally or remotely); c) Grounding of a third lead shall indicate when the STANDBY-DEFRUIT-REMOTE switch is in REMOTE position; d) a fourth lead shall indicate the power ON-OFF condition. Power (6.3 V AC for these light indicators located up to 91.5 m (300 ft) from the defruiter) shall be provided by the defruiter. FIGURE 1 is a recommended guide for accomplishing the remote control requirements.

3.3.2.17 Traffic capacity. The defruiting equipment shall be capable of both pulse-to-pulse and bracket defruiting the composite video signals when pulses of one reply code group are interleaved with those of another. The definition of interleaving precludes overlapping (see 6.5.6). Under the conditions of maximum duty cycle specified herein, the output signal amplitude for a constant input signal amplitude shall not deviate by more than 1 dB from that obtained when there is only one reply code group for each interrogation.

3.4 Workmanship. Workmanship shall conform to Requirement 9 of MIL-STD-454.

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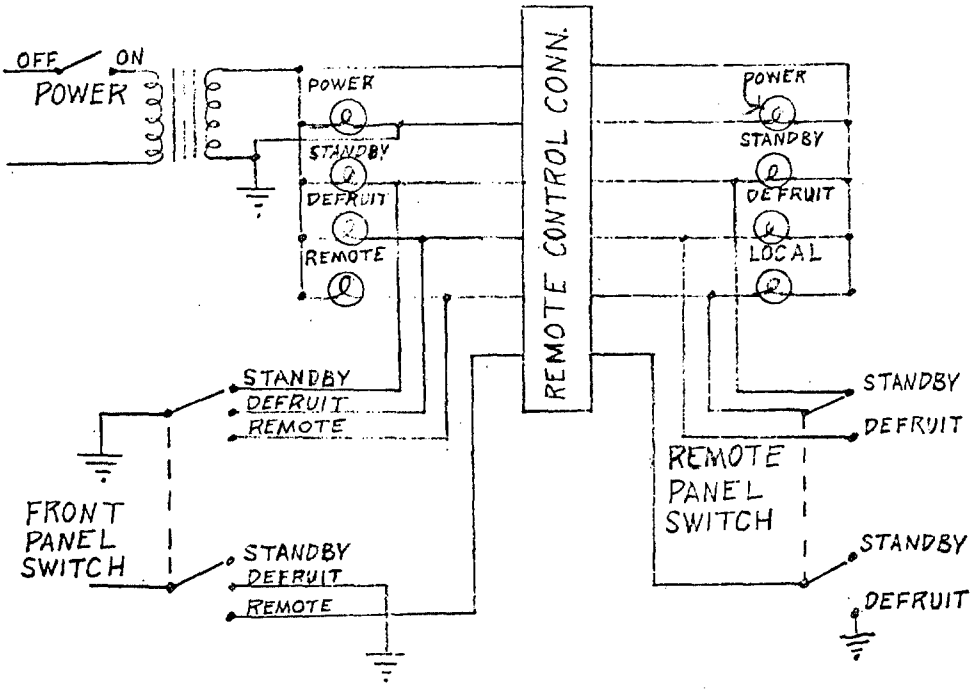


FIGURE 1. Remote control operation.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.3 Inspection procedures. Unless otherwise specified herein, inspection procedures shall be in accordance with MIL-E-16400F.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (see 4.4.1)
   2. Production control inspection (4.4.2)
   3. Environmental inspection (4.4.3)
c. Reliability demonstration (see 4.5)
d. Maintainability demonstration (see 4.6)
e. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified in the contract (see 6.2.1) two sample unit(s) of the MX-8757/UPX and two sample units of the MX-8758/UPX shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the entire preproduction inspection requirements of MIL-E-16400F and the additional tests specified in 4.3.2, 4.3.3, 4.3.4, and 4.3.5. Water cooling, accelerated life, noise and reliability tests specified in MIL-E-16400F are not required.

4.3.1 Operating test (first article inspection). The operating test shall be in accordance with MIL-E-16400F and shall include the tests specified in 4.3.1.1, 4.3.1.2, and 4.3.1.3.

4.3.1.1 Test of traffic capacity. A test of traffic-handling capability will be the successful processing of clusters of eight interleaved reply code groups (code groups to contain an average of 10 pulses each) in a period of 120 μs with clusters repeated 10 times during the 4000 µs interrogation period. These reply code groups shall be synchronous with the interrogator period. These reply code groups shall be synchronous with the interrogator pre-trigger. At the same time random pulses at the rate of 100,000 pps shall be introduced into the defruiter and noise shall be introduced into the defruiter input to produce a duty factor (signal pulse noise) at the defruiter input of 25 percent. For purposes of this test, the coded reply and random pulse input signals shall be composed of pulses 0.45 μs duration of 2.0 volt input level.

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4.3.1.2 Test of signal-to-noise ratio. A test of the signal-to-noise ratio shall be conducted as follows: Simulated tags and reset signals shall be supplied the defruiter at the rate of 250 sets per second. One pulse generator shall be synchronized with these triggers and shall generate pulses at the rate of 50,000 per second to simulate synchronous replies. A second pulse generator shall be run to the first asynchronously (triggered by a noise generator) and shall generate pulses at the rate of 60.000 per second to simulate fruit. With the signal-to-noise ratio (as specified in 3.3.2.4.2 c) equal to two or greater, the defruiter output shall contain a maximum of 3000 fruit pulses per second.

4.3.1.3 Test of video fidelity. The fidelity and recovery characteristics of the video circuitry shall be such that, with an input noise level of 0.5 volt root-mean-square (rms), an output limit level set to 2.0 volts, and with two input pulse trains(first pulses separated by 0.7 μs) as specified herein, the outputs of these pulse trains across a 75 ohm external load are as specified in 4.3.1.3.1 through 4.3.1.3.4.

4.3.1.3.1 Input train number 1. The input train number 1 requirements are as follows:

a. Number of pulses: 15
b. Code: Code 7777 plus SPI pulses in accordance with DoD AIMS 65-1000 Technical Standard for reply pulse spacing.
c. Duration: 0.55 μs (+ 0.00 μs, -0.05 μs)
d. Amplitude: Start pulse 2.0 V $\pm$ 0.1 V
e. Rise time: 0.1 μs, maximum
f. Decay time: 0.2 μs, maximum
g. Droop amplitude of pulse train: Not greater than 1 dB

4.3.1.3.2 Input train number 2. The input train number 2 requirements are as follows:

a. Number of pulses; 15
b. Code: Code 7777 plus SPI pulse in accordance with DoD AIMS 65-1000 Technical Standard for reply pulse spacing.
c. Duration: 0.35 μs $\pm$ 0.02 μs
d. Amplitude: Start pulse 0.75 V $\pm$ 0.1 V
e. Rise time: 0.1 μs, maximum
f. Decay time: 0.2 μs, maximum
g. Droop amplitude of pulse train: Not greater than 1 dB

4.3.1.3.3 Output pulse train produced by input train number 1. The output pulse train produced by input train number 1 is as follows:

a. Number of pulses: 15
b. Code: Same as input number 1
c. Duration: Input pulsewidth $\pm$ 0.03 μs
d. Amplitude: 2.0 V $\pm$ 0.1 V
e. Rise time: Input rise time $\pm$ 10 percent
f. Decay time: Input decay time $\pm$ 10 percent
g. Droop amplitude of pulse train: Input droop + 1 dB, maximum
h. Pulse spacing: Spacing of any pulse referenced to start pulse of the output train shall not differ from that of the input train by more than $\pm$ 0.02 μs.

4.3.1.3.4 Output pulse train produced by input train number 2. The output pulse train produced by input train number 2 is as follows:

a. Number of pulses: 15
b. Code: Same as input number 2
c. Duration: Input pulsewidth $\pm$ 0.03 μs
d. Amplitude: 2.0 V $\pm$ 0.1 V

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4.3.2 Altitude (elevation) tests. Altitude (elevation) tests shall be conducted in accordance with MIL-STD-810, Method 500, Procedure 1, except that the operating test shall be performed at 3658 m (12,000 ft).

4.3.3 Sand and dust tests. Sand and dust tests shall be conducted in accordance with MIL-STD-810, Method 510, except that the dust used shall have a size distribution of particles 10 to 50 microns.

4.3.4 Leakage current test

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITON OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.3.4.1 Equipment connections. Each MX-8757/UPX and MX-8758/UPX unit shall be separately placed on an insulated surface. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 2.

4.3.4.2 Measurement. Leakage current shall be measured on equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage shall be measured across the 1500 ohm resistor at 115 VAC source power and at both 60 Hz and 400 Hz. A measurement of 7.5 volts or greater as determined by the voltage drop method shall correspond to a current of 5 mA or greater. The overall measurement error shall not exceed five percent. The probe shall be used on all externally accessible conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. The open safety ground conductor shall be reconnected immediately after the test is completed.

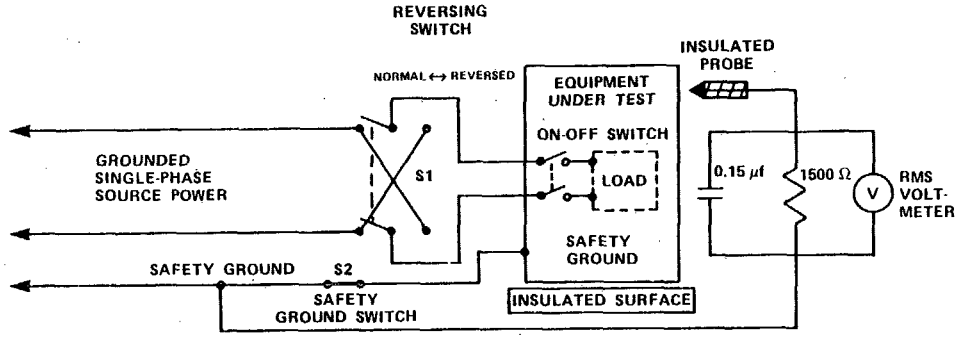
4.3.5 Interchangeability demonstration. The subassemblies and plug-in assemblies of the Interference Blankers shall be interchanged between the first article units to demonstrate conformance to the physical and functional interchangeability requirements of 3.2.15.6. The operating test of 4.3.1 shall be performed after interchanging subassemblies and plug-in assemblies.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Unless otherwise specified herein, the production inspection shall be in accordance with MIL-E-16400 and shall be conducted on every equipment offered for delivery.

4.4.1.1 Operating test (production inspection). The operating test required by MIL-E-16400 shall include the tests specified in 4.3.1.1, 4.3.1.2, and 4.3.1.3 and shall be performed at an elevated temperature of 50°C (+5°C, -0°C) following a two-hour stabilization period at this temperature. The duration of the test shall be a cumulative operating time of 24 hours of failure-free time (see 4.4.1.1.1). No adjustment of the equipment, other than normal operating mode changes, shall be allowed during the test. Failures occurring during the 24 hours shall be repaired and the operating test resumed until 24 hours of failure-free time is achieved.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW reversed. ON-OFF SW ON. REPEAT STEP 4.
7. ON-OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 2. Single-phase test diagram for leakage current measurement.

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4.4.1.1.1 Failure. Failure is any malfunction or parameter deviation that prevents the euqipment from performing within the operational requirements specified herein.

4.4.2 Production control inspection. Production control inspection shall be in accordance with MIL-E-16400F and shall include leakage current measurements (see 4.3.4). Electromagnetic interference emission and susceptibility and equipment and systems self-compatibility, the weld test, and reliability tests are not required. Production control inspection shall be in accordance with the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection, with an acceptable quality level (AQL) of 4.0 percent. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified in the contract (see 6.2.1), sample units which have been subjected to and have passed production control inspections may be accepted on the contract, provided they are resubjected to and pass production inspection, after repair of all visible damage.

4.4.3 Environmental inspections. A sample of one out of every group of 75 equipments, or fraction thereof, which have passed the production inspection tests of 4.4.1 shall be selected and subjected to the following environmental tests of MIL-E-16400F:

a. Temperature
b. Humidity
c. Shock
d. Vibration
e. Salt spray

A sample of one unit out of every group of 200 units, or fraction thereof, which have passed the production inspection tests, shall be selected and subjected to the electromagnetic interference emission and susceptibility and equipment and systems self-compatibility tests of MIL-E-16400F.

4.4.3.1 Sampling for environmental inspection. Sample units for the first environmental test shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the Quality Assurance Representative does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract (see 6.2.1), sample units which have been subjected to and have passed the environmental tests of 4.4.3 may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Reliability demonstration. The reliability demonstration shall be performed in accordance with MIL-STD-781.

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4.5.1 Reliability qualification test. A sample of at least two randomly selected units from early production delivery units shall be subjected to reliability qualification testing in accordance with Test Plan IV C of MIL-STD-781 to demonstrate a lower test MTBF (θ_1) of 1000 hours. Environmental and operational duty cycles shall be as specified herein:

a. Temperature: $50^{o}C$, $+5^{o}C$, $-0^{o}C$
b. Moisture: Once per test day, moisture shall be injected into the test chamber during the stabilized hot temperature period. The moisture level need not be strictly controlled, but shall be sufficient to cause visible condensation in the chamber.
c. Operating duty cycle: Turn on and let temperature stabilize, hold for the remainder of a 24-hour day, then turn off and let temperature stabilize. This cycle shall continue throughout the test.
d. Input voltage: Nominal specified voltage + 5 percent, -2 percent.

4.5.1.1 Failure. The procuring activity shall be notified within two working days of any failure during reliability testing. Classification of failures shall be made by the procuring activity in accordance with the Failure actions paragraph of MIL-STD-781.

4.5.1.2 Rejected equipment. In the event an equipment is rejected as a result of reliability testing, the contractor shall immediately notify the procuring activity of the cause(s) for such rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the procuring activity and retrofitted to all production units, the reliability qualification test shall be reperformed as specified in 4.5.1.

4.5.2 Equipment conditioning. Each equipment and spare module shall conform to the equipment conditioning requirement of 3.2.3.1.2. The requirement for equipment conditioning shall be considered as having been satisfied upon successful performance of the operating test as required under production inspection (see 4.4.1.1).

4.6 Maintainability demonstration. Conformance to the maintainability requirements of 3.2.3.2 shall be demonstrated as specified in Method 4 of MIL-STD-471. The maintainability demonstration shall be successfully completed prior to delivery of any production equipment.

4.7 Workmanship. Inspections shall be conducted to ensure conformance to the requirements of 3.4

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The Interference Blankers MX-8757/UPX and MX-8758/UPX are intended for use on board ships and in shore installations.

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6.2 Ordering data

6.2.1 Procurement requirements. Procurement documents should specify the following;

a. Title, number, and date of this specification
b. Types of input and output connectors if other than specified in 3.2.12.3
c. Number of first article samples required of other than specified in 4.3
d. When production control sample units may not be accepted (see 4.4.2.2) (see 4.4.2.2)
e. When environmental sample units may not be accepted (see 4.4.3.3)
f. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contractor requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.1.3	Test procedures	UDI-T-11710
4.3	First article inspection report	UDI-R-22576A
4.5	Reliability demonstration test plans	DI-R-2124
4.5.1	Reliability test reports	DI-R-2118
4.6	Maintainability demonstration plans	DI-R-2129
4.6	Maintainability demonstration report	DI-R-2130A

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first articles should be first production items. The first article should consist of two sample units of the MX-8757/UPX and two sample units of the MX-8758/UPX. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

6.4 Definitions. Definitions of terms concerning the general characteristics of the information employed or processed by the defruiter equipment covered by this specification are given in 6.4.1 through 6.4.20.

6.4.1 Code structure. The IFF code structure is as specified in the DoD AIMS 65-1000.

6.4.2 Defruit. Interference suppression (or blanking) or deletion of random, non-synchronous replies from the video input to the decoding/processing equipment by comparing video reply signals on two successive IFF interrogations on the same mode.

6.4.3 Pulse-to-pulse defruiting. A type of defruiting which compares each pulse in a received reply with the corresponding pulse (or pulse position) received on the previous reply (on the same mode) to determine time synchronism. When an undelayed pulse in a received reply is in time synchronism with a pulse in the delayed reply, the undelayed pulse is passed to the output.

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6.4.5 Code garbling. A form of signal interference in a pulse train, caused by the overlapping (excluding interleaving) of two or more trains, or by the presence of one or more pulses in any of the openings among nominal pulse positions during or within one train length from legitimate signal train. The effect may be false code information or cancellation of a desired code, as when a reply pulse from another transponder occurs at a vacant pulse position in the code train from the desired transponder.

6.4.6 Interleaving. Two or more pulse trains whose time positions are such that the pulses of one train occur during the interval between pulses of the other trains.

6.4.7 Mode identification tags. A series of pulses used to indicate the mode of interrogation. Interrogation pulses are used as mode identification tags and occur prior to zero range for each interrogation.

6.4.7.1 Characteristics. Mode identification tags consist of the following:

a. Mode 1 — 2 pulses 0.8 (μs) $\pm$ 0.1 μs wide, spaced 3 μs $\pm$ 0.2 μs apart. Amplitude between 2.0 volts (V) and 20.0 V.

b. Mode 2 — 2 pulses 0.8 μs $\pm$ 0.1 μs wide, spaced 5 μs $\pm$ 0.2 μs apart. Amplitude is between 2.0 V and 20.0 V.

c. Mode 3/A — 2 pulses 0.8 μs $\pm$ 0.1 s wide spaced 8 μs $\pm$ 0.2 μs apart. Amplitude is between 2.0 V and 20.0 V.

d. Mode C — 2 pulses 0.8 μs $\pm$ 0.1 μs wide spaced 21 μs $\pm$ 0.2 μs apart. Amplitude is between 2.0 V and 20.0 V.

e. Mode 4 — 4 pulses 0.5 μs $\pm$ 0.1 μs wide spaced 2.0 μs $\pm$ 0.1 μs (as referenced to the first pulse), 4 $\pm$ 0.1 μs and 6 μs $\pm$ 0.1 μs.

f. Side lobe suppression pulse may be present in all five modes. For Modes 1, 2, 3/A, and C these pulses occur 2 $\pm$ 0.1 μs after the first pulse and are 0.8 $\pm$ 0.1 μs wide. For Mode 4 the pulse occurs 8 μs $\pm$ 0.1 μs after the first pulse and is 0.5 $\pm$ 0.1 μs wide. Amplitude is between 2.0 and 20.0 volts.

6.4.8 Reset signal. The reset signal is that identification signal which indicates the end of the receiver gate or the end of IFF ranging time. The reset signal consists of three pulses. The first pulse is 8.0 $\pm$ 0.25 μs wide and occurs 2.0 to 3.0 μs after the end of the receiver gate. The second pulse, 2.0 $\pm$ 0.25 μs wide, occurs 10.0 $\pm$ 0.25 μs after the leading edge half voltage point of the first pulse. The third pulse is 4.0 $\pm$ 0.25 μs wide and occurs 15.0 $\pm$ 0.25 μs after the leading edge half voltage point of the first pulse. The end of the receiver gate is defined as that point in which the output video has decreased 24 dB below normal but no later than 3.0 μs after the video has decreased to 24 dB. The amplitude of the reset signals shall be between 2.0 and 20.0 volts.

6.4.9 Separated video. That system of IFF video in which each mode of video is provided on a single separate video line.

6.4.10 Composite video. That system of IFF video in which all modes of video are provided on the same single line which is different from the line or lines on which mode identification information occurs.

6.4.11 Mode identification (composite) tags and reset signals. That system of tag and reset signals in which the tag and reset signals both occur on the same line but separate from the video line or lines.

6.4.12 Tagged video. A system of composite video to which has been added (on the same line) mode identification tags and reset signals. The tags and reset signals shall occur for each interrogation.

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6.4.13 Mode sensitive triggers. A system of mode identification in which a single trigger pulse is provided on a separate line for each mode of operation occuring at zero range time. Trigger duration is not less than 0.5 μs and amplitude is from 2.0 V to 20.0 V.

6.4.14 IFF pretrigger. A pulse generated by the associated radar set or IFF pretrigger generator and provided to the IFF system in advance of the radar main trigger to initiate the generation of an IFF interrogation.

6.4.15 Signal-to-noise ratio. The signal-to-noise ratio is defined as:

$$\frac{\text{Signal (peak) + noise (rms)}}{\text{Noise (rms)}}$$

6.4.16 Undelayed video. Real-time video (see FIGURE 3).

6.4.17 Delayed video. The video which has been stored in memory (see FIGURE 3).

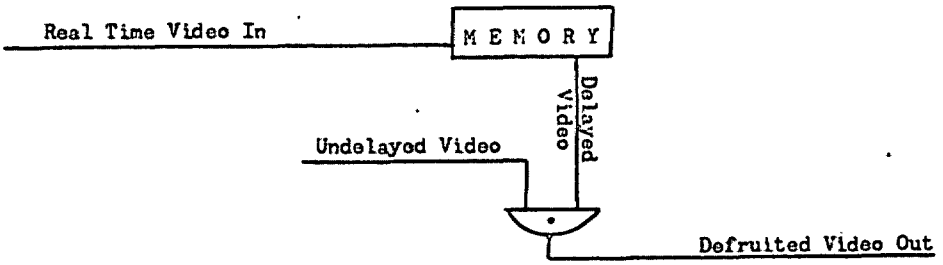


FIGURE 3. Video memory storage.

6.4.18 Acceptance gate. The gate which is generated as a result of the pulse position of the pulses in the delayed video pulse train which allows any video from the undelayed video pulse train falling within that gate to be passed.

6.4.19 Fruit. IFF reply pulses received in response to interrogations from other IFF interrogation sites. IFF reply pulses received at an IFF interrogation site which are not time synchronous with the site's IFF system.

6.4.20 Channel. A single chain of mode dependent memory.

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